THE CITY COLLEGE OF NEW YORK · BIOMEDICAL ENGINEERING

BME i9400
Special Topics in Machine Learning

Fall 2026

INSTRUCTOR	Jacek Dmochowski, PhD · jdmochowski@ccny.cuny.edu	OFFICE	Steinman Hall 460 · hours TBA
MEETINGS	Mon & Wed 11:00–12:15 · 28 total · Steinman Hall C-51	CREDITS	3
RUNTIME	Google Colab	PLATFORM	Brightspace · github.com/BME-i9400

ABOUT THIS COURSE

This is a hands-on graduate course in biomedical machine learning. You will learn to frame a biomedical question as a learning problem, build a defensible baseline, adapt a modern model, and evaluate the result with the rigor that biomedical use demands.

It is deliberately not an encyclopedic survey of algorithms, nor a course in training frontier models from scratch. Implementation is easy now. What remains hard — and what this course is about — is data readiness, leakage-free evaluation, model choice, calibration, and interpreting what the model learned.

Prerequisites: basic Python, and prior exposure to probability/statistics and linear algebra. No previous machine learning course is required.

> **Generative AI is permitted and expected on out-of-class work.** That is how you will work in your career. What is graded is not the artifact but your ability to defend it, in class.

LEARNING OUTCOMES

By the end of the course you will be able to:

1. Frame a biomedical question as a prediction task, stating explicitly what a row of X is and what y is.
2. Build reproducible, leakage-free train/validation/test workflows and spot the data-quality risks that threaten them.
3. Select metrics and operating thresholds suited to the use case, accounting for imbalance and prevalence.
4. Build, regularize, and interpret linear-model and tree-ensemble baselines.
5. Train neural networks in PyTorch and adapt pretrained image, signal, and text models rather than treat them as black boxes.
6. Determine what a model learned via coefficient, permutation, and gradient-based attribution.
7. Determine whether an effect is real via bootstrap intervals and paired model comparison.

HOW THE COURSE RUNS

Two meeting types, planned as alternating but will likely not always be due to time restrictions.

Concept meeting. Conventional lectures, sometimes including one or more short quizzes (tentatively planned to be delivered over Brightspace.)

Studio meeting. Twenty minutes of targeted instruction followed by forty-five minutes of in-class Python notebook assignments.

ASSESSMENT

COMPONENT		NOTES
In-class quizzes	10%	About 15 across the semester, scored 0/1/2; your best 10 count.
Labs & defense quizzes	20%	Three labs, released and defended roughly two weeks apart. AI is permitted on the lab. Each is defended by a ten-minute, closed-AI, in-class quiz worth 70% of the component.
Midterm exam	20%	**Wednesday, October 28.** 75 minutes, closed book, pencil and paper exam.
Final concept exam	15%	**Final examination period, Dec 15–21.** 45 minutes, closed book, pencil and paper exam.
Capstone project	35%	An individual end-to-end study built from an instructor-supplied topic. Oral presentation and live Q&A 20%; written report 15%. Full details released during the course.
Total	100%	

There are no conventional graded homework sets. The three labs are graded primarily through in-class defense, not through the submitted code.

POLICIES

Attendance. This is a studio course; the work happens in the room. Attendance is not graded directly, but quiz credits can only be earned in class. Your five dropped quizzes are the absence allowance — use them for illness, conferences, and interviews. Please do not email after the fact to explain a missed meeting.

Late work. Lab notebooks may be up to 48 hours late with a 20% penalty on the notebook portion, but *defense quizzes cannot be made up* except for a documented excused absence. Capstone milestones lose 20% per day. Documented medical and emergency exceptions are handled case by case.

Exam conflicts. Notify the instructor at least one week in advance of any documented conflict with an exam date.

Academic integrity. Discussion of studio work is encouraged; submitted notebooks must be individually authored. Violations of the CCNY Academic Integrity Policy are reported and may result in course failure.

Accessibility. Students requiring accommodations should contact the AccessAbility Center (NAC 1/218) and notify the instructor early in the term.

MATERIALS

A laptop with a modern browser, a Google account for Colab, and a CUNY Brightspace account. Notebooks and curated public datasets are supplied. Colab is the required runtime and all graded work assumes it; a local Python setup is optional and unsupported.

SCHEDULE (SUBJECT TO CHANGE)

#	DATE	TYPE	TOPIC	MILESTONE
1	Mon 8/31	ASYNCHRONOUS	Course launch, Colab setup, and biomedical data readiness: provenance, label quality, splits, leakage	
2	Wed 9/2	CONCEPT	Probability for diagnosis: Bayes, prevalence, sensitivity/specificity, PPV/NPV, likelihood ratios	
3	Wed 9/9	STUDIO	Screening simulation; NumPy and pandas on-ramp	

#	DATE	TYPE	TOPIC	MILESTONE
4	Mon 9/14	CONCEPT	Linear algebra for ML: vectors, projection, SVD, PCA, embeddings	
5	Wed 9/16	STUDIO	PCA of a biomedical feature matrix	
6	Wed 9/23	CONCEPT	Regression: least squares, normal equations, the geometric view, the noise model	
7	Mon 9/28	STUDIO	Fit, regularize, and diagnose a regression baseline	**Lab 1 released**
8	Wed 9/30	CONCEPT	Linear classification I: logistic regression, cross-entropy, decision boundaries	
9	Mon 10/5	CONCEPT	Linear classification II: confusion matrix, ROC and PR curves, thresholds, imbalance	
10	Wed 10/7	STUDIO	An end-to-end clinical tabular classifier	**Lab 1 defense**
11	**Tue 10/13**	**CONCEPT**	Regularization and cross-validation: L1/L2, bias–variance, leakage inside CV	**Monday schedule**
12	Wed 10/14	STUDIO	Nested cross-validation and calibration	
13	Mon 10/19	CONCEPT	Optimization and generalization: GD/SGD, learning rates, loss surfaces, convergence	
14	Wed 10/21	STUDIO	Diagnosing underfit, overfit, and unstable training runs	**Lab 2 released**
15	Mon 10/26	REVIEW	Synthesis and midterm review	
16	**Wed 10/28**	**EXAM**	**Midterm examination**	
17	Mon 11/2	CONCEPT	Neural networks: the MLP, forward and backward pass, capacity, regularization	
18	Wed 11/4	STUDIO	A PyTorch MLP on clinical tabular data	**Capstone briefs released · Lab 2 defense**
19	Mon 11/9	CONCEPT	Trees, ensembles, and attribution: random forests, gradient boosting, permutation importance, SHAP	
20	Wed 11/11	STUDIO	Is the difference real? Bootstrap confidence intervals and paired model comparison	**Capstone brief selection due**
21	Mon 11/16	CONCEPT	Convolutional networks: convolution, receptive fields, pooling, transfer learning	
22	Wed 11/18	STUDIO	Fine-tuning a small image encoder on a curated biomedical task; Grad-CAM	**Lab 3 released**
23	Mon 11/23	COMBINED	Sequence models for biosignals: windowed features versus 1D-CNN and GRU on ECG/EEG	**Capstone analysis plan due**
24	Mon 11/30	COMBINED	Text, embeddings, and pretrained encoders; attention conceptually	
25	Wed 12/2	COMBINED	Biomedical LLM workflows: prompting, structured extraction with schema validation, auditing outputs	**Lab 3 defense**
26	Mon 12/7	CLINIC	Capstone clinic: baselines reproduced, results tables reviewed, peer critique	
27	Wed 12/9	PRESENT	Capstone presentations I	8 min + 4 min graded Q&A
28	Mon 12/14	PRESENT	Capstone presentations II	8 min + 4 min graded Q&A
–	**Dec 15–21**	**EXAM**	**Final concept exam** (45 min) and capstone presentations III	**Written package due**

All information in this syllabus is subject to change at the instructor's discretion.